Econometrics Part 1

The linear model: presentation

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Outline

1. Introduction
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4. Distribution and asymptotics
5. Goodness of fit
6. The multiple linear model
7. OLS in the multiple linear model
8. Geometry of OLS
9. Analysis of variance and the R^2

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Topics addressed

- Linear regression
- Heteroskedasticity and autocorrelation
- Instrumental Variables
- Simultaneous Equations
- Panel Data
- Discrete choice
- Censored variables
- Time series

Econometrics

- Economic analysis relies on theoretical representations of behaviors and mechanisms (e.g. macro-economic models, the Capital Asset Pricing Model, etc.)
- The relevance of these models should be tested on real data (e.g. does the CAPM "work"?)
- Parameters of interest should be assessed too (e.g. what is the impact of a rise in oil prices or interest rates on car sales?)
- Econometrics uses statistical tools to provide an answer to these questions
- Here, theoretical and hands-on econometrics: How to manage real data? How to find the best model? Did I get what I expected? Can I trust this model?

Example

Consider an individual consumption function: for each individual i, we observe consumption C_i and income I_i. A very simple model explaining C_i would be:

$$\forall i, \quad C_i = a + bI_i$$

The goal of econometrics would be to test the relevance of this model (is it a *good approximation* of reality?), to provide *estimates* of parameters a and b and to assess the accuracy of these estimates (inference: tests). Plus, one may be concerned that the link between C and I is not that perfect.

Types of datasets

- *Cross-sectional data*: individual data collected at a particular point in time (e.g. census data on individuals, households, firms)
- *Time series data*: variables collected over time (e.g. yearly GDP, monthly unemployment rate)
- *Panel data*: individual data collected over time (e.g. Medical Expenditure Panel Survey)

Cross-sectional data and panel data are collected not at the *population* level, but on a *sample*, representative of the population. We thus have to infer the characteristics of the population from the information we get from that particular sample.

Variables

- Quantitative variables are measured on a numeric or quantitative scale: income, age, ...
- Qualitative variables are not: gender, region, ..., but need to be coded with numbers anyway (*dummy variables*)

Consider model $\forall i,\ C_i = a + bI_i$

- C_i: the endogenous (*or* dependent *or* explained) variable, which is the one explained by the model
- I_i and the constant: exogenous (*or* independent *or* explanatory) variables that determine the endogenous variable

The simple linear model: example

Say we'd like to know about the health care habits of the French population. Assume that for each individual i, health care expenditure y_i could be explained by age x_i, in a linear way:

$$\forall i, \quad y_i = b_0 + b_1 x_i$$

For a newborn, expenditure would be b_0, and each additional year would incur b_1 extra cost. This makes sense, since we all know that health expenditure increases with age. This is a simple model because we have only one explanatory variable.

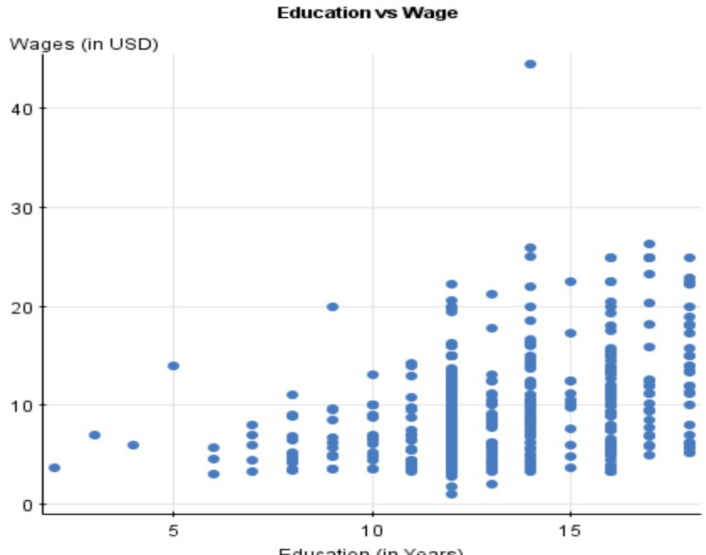
Error term - linearity

Of course, the link between age and expenditure cannot be that perfect, so we add an **error term** u_i that accounts for unobserved factors that make this relation not perfect (frailty, income, etc.).

$$\forall i, \quad y_i = b_0 + b_1 x_i + u_i$$

This is a (simple) model of individual health care expenditure. This is how we believe the data was generated in the first place. Its relevance cannot be assessed on the whole population, but rather on a sample, say of size N. The model can be estimated using ordinary least squares (OLS), if some assumptions are satisfied. Notice that this is a linear model because parameters (and not necessarily variables) enter it in a linear way.

Example


Education vs Wage
Wages (in USD)
40
30
20
10
0
5
10
15
Education (in Years)

The simple linear model: assumptions

Let's call i an individual from the size N sample.

$$\forall i, \quad y_i = b_0 + b_1 x_i + u_i$$

Calling E the expected value and V the variance, the usual and so-called *Gauss-Markov assumptions* for such a model are:

1. $\forall i,\ E(u_i|X) = 0$
2. u and x are uncorrelated
3. $\hat{V}(x) \neq 0$
4. $\forall i,\ V(u_i) = \sigma^2$
5. $\forall i \neq j,\ \text{cov}(u_i, u_j) = 0$

Everything is conditional on X; we don't write it every time for simplicity.

What they mean

1. We want the model to predict the right outcome, given any x:
   $\forall i,\ E(y_i|x_i) = E(b_0 + b_1 x_i + u_i|x_i) =$
   $E(b_0 + b_1 x_i|x_i) + E(u_i|x_i) = b_0 + b_1 E(x_i|x_i) = b_0 + b_1 x_i$
2. u should be totally random: no way to guess u from X
3. The x_i can't all be the same
4. The variance of the error term should be the same for all individuals of the sample (*homoskedasticity*)
5. The error terms of different individuals should be uncorrelated (*no autocorrelation*)

Reminder (1): E and V

- $E(X)$ is the *expected value* of X: theoretical mean
- $\bar{X}$ is the *average* of observed X: empirical mean
- $V(X)$ is the *variance* of X: how we expect X to vary around its mean (theoretical)
- $V(X) = E((X - E(X))^2)$
- $\hat{V}(X)$ is the *empirical variance* of X: how X indeed varies around its mean
- $\hat{V}(X) = \frac{\sum_{i=1}^{N}(X_i - \bar{X})^2}{N}$
- To get an unbiased estimator of $V(X)$, there is a minor adjustment: we divide by $(N-1)$ instead of N

Reminder (2): Conditioning

- Earlier, we wrote: $E(y_i|x_i) = E(b_0 + b_1 x_i + u_i|x_i)$
- Say y is income and x is gender: 0 for males, 1 for females
- Expected income for a male, i.e. knowing $x = 0$:
  $E(y_i|x_i = 0) = b_0 + b_1 \times 0 = b_0$
- Expected income for a female, i.e. knowing $x = 1$:
  $E(y_i|x_i = 1) = b_0 + b_1 \times 1 = b_0 + b_1$
- So b_0 is the expected male income, $b_0 + b_1$ is the expected female income, and b_1 is the expected difference
- $\Rightarrow E(y_i|x_i)$ is used to predict individuals' outcomes, given their characteristics

Reminder (3): Conditioning

- Without conditioning on x, this would be:
  $E(y_i) = E(b_0 + b_1 x_i + u_i) = b_0 + b_1 E(x_i)$
- $E(x_i)$ is the expected proportion of females in the population
- So $E(y_i)$ is the average income over the whole population, i.e. the weighted average of males' and females' incomes
- $\Rightarrow E(y_i)$ is used to predict the overall average outcome
- Usually, people discuss $E(y_i)$ while implicitly discussing $E(y_i|x_i)$ instead, because the latter is more informative

Interpretation of parameters

Say we have the following model:

$$\forall i, \quad y_i = a + bx_i + u_i$$

Then b can be interpreted as the marginal change in y when x increases by 1 unit since, thanks to assumption 1 ($E(u|x) = 0$), we have:

$$\frac{\partial E(y|x)}{\partial x} = b$$

The linear model is in fact more general than it looks (see the next slides).

Example: squared variables

The following model is linear, even if variable x is squared:

$$y_i = a + bx_i^2 + u_i$$

And a way to understand how y changes with x is to compute the following derivative (let's drop i's for convenience):

$$\frac{dy}{dx} = 2bx$$

And we see it is not constant: it depends on the value of x. The relationship between x and y is not a line but a curve. It is even better to put both x and x^2 in the equation to allow for more flexibility, which is what is done in practice:

$$y_i = a + bx_i + cx_i^2 + u_i \qquad \text{with} \qquad \frac{dy}{dx} = b + 2cx$$

Example: logs (1)

- Let's have a simple model: $\forall i,\ \log(y_i) = a + bx_i + u_i$
- How can we interpret b? (let's drop i's for convenience)
- Taking the exponential: $y = \exp(a + bx + u)$

$$\frac{dy}{dx} = b\exp(a + bx + u) = by$$

$$\frac{dy/y}{dx} = b$$

$100 \times b$ is thus interpreted as the **percentage** of variation of y when x increases by 1 unit (approximately, for small b). This kind of model has a *semi-log* form.

Example: logs (2)

- Let's have a simple model: $\forall i,\ \log(y_i) = a + b\log(x_i) + u_i$
- How can we interpret b?
- Taking the exponential: $y = \exp(a + b\log(x) + u)$

$$\frac{dy}{dx} = b\frac{1}{x}\exp(a + b\log(x) + u) = b\frac{y}{x}$$

$$\frac{dy/y}{dx/x} = b$$

b is thus interpreted as an **elasticity**: the % of variation of y when x increases by 1%. This kind of model has a *log-log* form.

How to choose?

- Using economic theory, common sense, plots, ...
- Example: the link between income and medical expenditures using the World Bank's 1997 Vietnam Living Standards Survey (source: Cameron & Trivedi, chapter 4)
- Model: $\textit{medical expenditures} = a + b \cdot \textit{income} + u$
- Data: 5,999 households
- Total household income is usually not well captured in developing countries, so we use total household expenditure as a *proxy*

Our goal: estimate the parameters of the model

$$\forall i, \quad y_i = b_0 + b_1 x_i + u_i$$

- b_0 and b_1 are unknown
- We thus want to find the best estimates for them: $\hat{b}_0$ and $\hat{b}_1$
- One way of doing this is by using ordinary least squares: OLS
- $\hat{b}_0 + \hat{b}_1 x_i$ should be as close as possible to y_i
- We define $\hat{y}_i$ as the model prediction of y_i: $\hat{y}_i = \hat{b}_0 + \hat{b}_1 x_i$
- And we define $\hat{u}_i$ as $y_i - \hat{y}_i$: the residuals
- Warning: u (unobserved error terms) and $\hat{u}$ (residuals) are two different things

OLS estimators

- We want the residuals to be as small as possible overall: this will give us the optimal $\hat{b}_0$ and $\hat{b}_1$
- But the residuals are known only once the regression line is drawn, which requires $\hat{b}_0$ and $\hat{b}_1$: so we express the residuals as a function of $\hat{b}_0$ and $\hat{b}_1$
- Since residuals can be either positive or negative, we minimize the sum of **squared** residuals with respect to $\hat{b}_0$ and $\hat{b}_1$:

$$\min_{\hat{b}_0,\hat{b}_1} \sum \hat{u}_i^2 = \min_{\hat{b}_0,\hat{b}_1} \sum (y_i - \hat{y}_i)^2 = \min_{\hat{b}_0,\hat{b}_1} \sum (y_i - \hat{b}_0 - \hat{b}_1 x_i)^2$$

And the solution is (proof below):

$$\hat{b}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \qquad \text{and} \qquad \hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x}$$

Remark

$$\hat{b}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{s_{xy}}{s_{xx}}$$

- $s_{xx} = \frac{1}{N}\sum(x_i - \bar{x})^2$ is the empirical variance of x
- $s_{xy} = \frac{1}{N}\sum(x_i - \bar{x})(y_i - \bar{y})$ is the empirical covariance between x and y
- So that:

$$\hat{b}_1 = \frac{\widehat{\text{cov}}(x, y)}{\hat{V}(x)}$$

- Warning: correlation $\neq$ causality

Proof (1)

Calling $S(\hat{b}_0, \hat{b}_1) = \sum \hat{u}_i^2$, we need the two First Order Conditions:

$$\frac{\partial S}{\partial \hat{b}_0} = 0 \Rightarrow -2 \sum_{i=1}^{N} (y_i - \hat{b}_0 - \hat{b}_1 x_i) = 0 \quad (1)$$

$$\frac{\partial S}{\partial \hat{b}_1} = 0 \Rightarrow -2 \sum_{i=1}^{N} x_i (y_i - \hat{b}_0 - \hat{b}_1 x_i) = 0 \quad (2)$$

Since the function is convex (strictly, as long as the x_i are not all equal) and twice differentiable, we don't really need the Second Order Conditions: we know in advance that the minimum exists and is unique.

Proof (2)

Equation 1 implies:

$$\sum y_i = N\hat{b}_0 + \hat{b}_1 \sum x_i \quad (3)$$

We divide everything by N, use the empirical means $\bar{y} = \frac{\sum y_i}{N}$ and $\bar{x} = \frac{\sum x_i}{N}$, and get:

$$\bar{y} = \hat{b}_0 + \hat{b}_1 \bar{x} \quad (4)$$

$\Rightarrow$ the regression line goes through the sample mean $(\bar{x}, \bar{y})$
(= the average person)
$\Rightarrow$ the average residual is always 0 (this is equation 1)

Proof (3)

Thanks to equation 4, we get $\hat{b}_0 = \bar{y} - \hat{b}_1\bar{x}$, and we plug in $\bar{y} - \hat{b}_1\bar{x}$ instead of $\hat{b}_0$ in equation 2:

$$\sum x_i(y_i - \bar{y} + \hat{b}_1\bar{x} - \hat{b}_1 x_i) = 0 \quad (5)$$

$$\sum x_i(y_i - \bar{y}) = \hat{b}_1 \sum x_i(x_i - \bar{x}) \quad (6)$$

Since $\sum \bar{x}(y_i - \bar{y}) = 0$ and $\sum \bar{x}(x_i - \bar{x}) = 0$, we can replace x_i by $(x_i - \bar{x})$ in front:

$$\hat{b}_1 = \frac{\sum x_i(y_i - \bar{y})}{\sum x_i(x_i - \bar{x})} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \quad (7)$$

The OLS estimator is linear

We have:

$$\hat{b}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

So:

$$\hat{b}_1 = \frac{\sum(x_i - \bar{x})y_i}{\sum(x_i - \bar{x})^2} = \sum \frac{(x_i - \bar{x})}{\sum(x_i - \bar{x})^2} y_i$$

Estimate $\hat{b}_1$ is thus a linear combination of elements y_i (with weights that only depend on the x's), and can also be written as a linear combination of elements u_i, since $y_i = b_0 + b_1 x_i + u_i$.

Are all observations given the same weight?

We have:

$$\hat{b}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

So:

$$\hat{b}_1 = \frac{\sum(x_i - \bar{x})^2 \frac{(y_i - \bar{y})}{(x_i - \bar{x})}}{\sum(x_i - \bar{x})^2} = \sum p_i \frac{(y_i - \bar{y})}{(x_i - \bar{x})}$$

Calling $p_i = \frac{(x_i - \bar{x})^2}{\sum(x_i - \bar{x})^2}$.
$\frac{(y_i - \bar{y})}{(x_i - \bar{x})}$ is the slope of the line drawn from the point corresponding to individual i to the sample average, and p_i is an increasing function of $|x_i - \bar{x}|$.
Estimate $\hat{b}_1$ is thus highly influenced by points with extreme x values (see Anscombe's quartet: four datasets with the same regression line).

OLS estimators are "BLUE"

$$\forall i, \quad y_i = b_0 + b_1 x_i + u_i$$

- OLS estimators are the Best Linear Unbiased Estimators (BLUE): this is the *Gauss-Markov theorem*, which is valid under the *Gauss-Markov assumptions* seen earlier
- They are linear in y_i, as seen earlier
- $\hat{b}_0$ and $\hat{b}_1$ are particular outcomes of random variables (their expression comprises y, which in turn comprises u), so we can compute their expected value and variance
- Unbiased: $E(\hat{b}_0) = b_0$, $E(\hat{b}_1) = b_1$
- "Best" means they have a minimal variance among unbiased linear estimators
- All this is true whatever the size of the sample
- Proof: in the general case with more than one x

The variance of estimators: summary

- Recall that $V(u_i) = \sigma^2$
- $V(\hat{b}_1) = \frac{\sigma^2}{\sum(x_i - \bar{x})^2}$, $\quad V(\hat{b}_0) = \frac{\sigma^2}{N} + \bar{x}^2 V(\hat{b}_1)$, $\quad \text{cov}(\hat{b}_0, \hat{b}_1) = -\bar{x} V(\hat{b}_1)$
- To get an estimate of these variances, we need an unbiased estimator of σ^2: $\hat{\sigma}^2 = \frac{\sum \hat{u}_i^2}{N-2}$
- Indeed, $\hat{V}(\hat{u}) = \frac{\sum(\hat{u}_i - \bar{\hat{u}})^2}{N} = \frac{\sum \hat{u}_i^2}{N}$
- And we need a small correction to make it an unbiased estimator of $V(u)$: divide by $N-2$ instead of N
- This is because we lose 2 degrees of freedom, since there are 2 parameters to estimate in the model (proof in the general case)

OLS estimators: summary

In model $y_i = b_0 + b_1 x_i + u_i$:

$$\hat{b}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

and

$$\hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x}$$

- Unbiased: $E(\hat{b}_0) = b_0$ and $E(\hat{b}_1) = b_1$
- Smallest variance among linear unbiased estimators: $V(\hat{b}_1) = \frac{\sigma^2}{\sum(x_i - \bar{x})^2}$ where $\sigma^2 = V(u_i)$
- Sensitive to extreme values

OLS estimators: distribution

- OLS estimators are a linear combination of u_i's
- If we had a large sample, we could rely on the Central Limit Theorem to learn about the distribution of estimators (see below)
- But for any sample size, the only way to know the distribution of estimators is to assume we know the distribution of the u_i's
- A common assumption is that $u_i \hookrightarrow N(0, \sigma^2)$
- It implies that both OLS estimators follow a normal distribution: $\hat{b}_0 \hookrightarrow N(b_0, \sigma^2_{\hat{b}_0})$ and $\hat{b}_1 \hookrightarrow N(b_1, \sigma^2_{\hat{b}_1})$
- Next, we'll check how these estimators behave with a large sample size

Asymptotics: a preview

- Do OLS estimators have good properties when the sample size goes to infinity, i.e. "asymptotically"?
- Let's call $(\hat{b}_{0,N})$ and $(\hat{b}_{1,N})$ the sequences of estimators computed on a sample of size N
- $(\hat{b}_{0,N})$ and $(\hat{b}_{1,N})$ *converge in probability* towards b_0 and b_1
- Definition of convergence in probability: (X_t) converges in probability towards a if and only if:
  $\forall \varepsilon > 0,\ P(|X_t - a| > \varepsilon) \to 0$ when $t \to +\infty$
- It means that when the sample size goes to infinity, the probability that the estimates are further than any given distance ε from the true parameters goes to 0

Additional assumption needed (1)

To prove this convergence, we need to assume that variable x follows this rule:

$$\lim_{N\to\infty} \frac{1}{N} \sum (x_i - \bar{x})^2 = \sigma_x^2 \neq 0$$

Which simply means that the empirical variance of x should have a given limit which is not zero. If it were zero, then it would mean that when we increase the sample size, after a while variable x wouldn't vary anymore (it would stick to its average $\bar{x}$), so additional x's wouldn't provide any information.

Additional assumption needed (2)

- In fact, the only thing we need here is that the x's keep some variance when the sample size goes to infinity, which is very likely to happen
- We also need the variance of x not to go to infinity, which is related to the *stationarity*[1] of the x's, so sometimes a weaker condition is used: $\lim_{N\to\infty} \sum(x_i - \bar{x})^2 = \infty$
- This weaker condition is enough for consistency, since $V(\hat{b}_1) = \frac{\sigma^2}{\sum(x_i-\bar{x})^2} \to 0$

[1] This part of the course focuses on micro data; stationarity is related to Time Series, covered later on

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Consistency

- Convergence in quadratic mean $\Rightarrow$ Convergence in probability
- Definition: a sequence of random variables (X_t) converges in quadratic mean towards a iff:
- $E[(X_t - a)^2] \to 0$ when $t \to \infty$
- Definition: (X_t) converges in probability towards a iff:
- $\forall \varepsilon > 0$, $P(|X_t - a| > \varepsilon) \to 0$ when $t \to \infty$

It means that we only need to prove that OLS estimators converge in *quadratic mean* towards their "true counterparts" to prove they are consistent. Since they are unbiased, $E[(\hat{b}_1 - b_1)^2] = V(\hat{b}_1)$: we only need to prove that their variance goes to zero when the sample size goes to infinity (easier).

Asymptotics

- Estimators are consistent, but can we know their distribution?
- Since estimators depend on error terms, if we don't know the distribution of the error terms, we won't know the distribution of estimators in finite samples
- But estimators are a linear combination of the error terms: thanks to the CLT[2], $\hat{b}_0$ and $\hat{b}_1$ are asymptotically normal
- So if the sample size goes to infinity (= with a sample "large enough"), we know the distribution of estimators, so we may compute confidence intervals, run tests, etc.

[2] Central Limit Theorem

The Central Limit Theorem (CLT)

(Lindeberg-Feller version)

- Assume we have a sequence of independent random variables $(z_1, z_2, \ldots, z_N)$ with finite expectations and variances: $\forall i,\ E(z_i) = \mu_i$ and $V(z_i) = \sigma_i^2$
- They can have different expectations and variances, and don't need to share the same distribution
- Call $\bar{z}_N = \frac{1}{N}\sum_{i=1}^{N} z_i$ and $\bar{\mu}_N = \frac{1}{N}\sum_{i=1}^{N} \mu_i$
- Assume $\bar{\sigma}_N^2 = \frac{1}{N}\sum_{i=1}^{N} \sigma_i^2$ and $\lim_{N\to\infty} \bar{\sigma}_N^2 = \bar{\sigma}^2 > 0$
- Plus a technical condition (Lindeberg's condition): no single z_i dominates the total variance
- Then: $\sqrt{N}(\bar{z}_N - \bar{\mu}_N)$ converges in distribution towards $N(0, \bar{\sigma}^2)$, when $N \to \infty$

Asymptotic results

Using the CLT and focusing on parameter $\hat{b}_1$, we get:

- $\sqrt{N}(\hat{b}_1 - b_1) \xrightarrow{d} N\left(0, \frac{\sigma^2}{\sigma_x^2}\right)$
- σ^2 is the variance of error terms u
- σ_x^2 is the asymptotic variance of x
- "Inflating" by $\sqrt{N}$ is needed, otherwise the limit would be degenerate (zero variance)

Summary: distribution of estimators

- When the sample size is large, OLS estimators become (approximately) Normal thanks to the Central Limit Theorem (CLT) if error terms are *iid*, whatever their distribution, because estimators are a linear combination of error terms
- When the sample size is not so large, estimators are Normal *only if* error terms are normal themselves, because these estimators are a linear combination of the error terms
- Knowing the distribution of estimators helps to draw inference (confidence intervals, tests, etc.)

A measure of the "goodness of fit" of the regression

We call the R^2 or *coefficient of determination*:

$$R^2 = \frac{\sum(\hat{y}_i - \bar{y})^2}{\sum(y_i - \bar{y})^2}$$

- It is the ratio of the *variance of the outcome explained by the model* over the *total variance of the outcome*, and lies between 0 (very poor fit) and 1 (perfect fit)
- Formal derivation of the R^2 will be given in the more general case of the multiple linear model, as well as the conditions under which it can be used
- Warning: if the x variable is a good predictor of y, but if the link is not linear, the R^2 will be low because of the lack of fit and not because of the poor choice of variables

The multiple linear model

Say we get back to the simple model explaining health care expenditures by age, and we want to add various other explanatory variables to the model (income, household size, etc.):
$x_2, x_3, \ldots, x_{k-1}$

$$\forall i, \quad y_i = b_0 + b_1 x_{i,1} + b_2 x_{i,2} + b_3 x_{i,3} + \cdots + b_{k-1} x_{i,k-1} + u_i \quad (8)$$

This is a (less simple) model of individual health care expenditure. It can be rewritten the following way, for individuals $i = 1$ to N:

$$\begin{aligned}
y_1 &= b_0 + b_1 x_{1,1} + b_2 x_{1,2} + b_3 x_{1,3} + \cdots + b_{k-1} x_{1,k-1} + u_1 \\
y_2 &= b_0 + b_1 x_{2,1} + b_2 x_{2,2} + b_3 x_{2,3} + \cdots + b_{k-1} x_{2,k-1} + u_2 \\
&\vdots \\
y_N &= b_0 + b_1 x_{N,1} + b_2 x_{N,2} + b_3 x_{N,3} + \cdots + b_{k-1} x_{N,k-1} + u_N
\end{aligned}$$

Multiple linear model: matrix notation

This system of equations can be rewritten using simple vectors:

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix} = b_0 \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} + b_1 \begin{pmatrix} x_{1,1} \\ x_{2,1} \\ \vdots \\ x_{N,1} \end{pmatrix} + \cdots + b_{k-1} \begin{pmatrix} x_{1,k-1} \\ x_{2,k-1} \\ \vdots \\ x_{N,k-1} \end{pmatrix} + \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_N \end{pmatrix} \quad (9)$$

And also in a more compact way, using matrices:

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix} = \begin{pmatrix} 1 & x_{1,1} & x_{1,2} & \cdots & x_{1,k-1} \\ 1 & x_{2,1} & x_{2,2} & \cdots & x_{2,k-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{N,1} & x_{N,2} & \cdots & x_{N,k-1} \end{pmatrix} \begin{pmatrix} b_0 \\ b_1 \\ \vdots \\ b_{k-1} \end{pmatrix} + \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_N \end{pmatrix} \quad (10)$$

Rewritten: $y_{(N,1)} = X_{(N,k)} b_{(k,1)} + u_{(N,1)}$

Some notations

For a random vector such as the estimator $\hat{b}$:

$$E(\hat{b}) = \begin{pmatrix} E(\hat{b}_0) \\ E(\hat{b}_1) \\ \vdots \\ E(\hat{b}_{k-1}) \end{pmatrix} \quad (11) \qquad V(\hat{b}) = E[(\hat{b} - E(\hat{b}))(\hat{b} - E(\hat{b}))'] \quad (12)$$

$$V(\hat{b}) = \begin{pmatrix} V(\hat{b}_0) & \text{cov}(\hat{b}_0, \hat{b}_1) & \cdots & \text{cov}(\hat{b}_0, \hat{b}_{k-1}) \\ \text{cov}(\hat{b}_1, \hat{b}_0) & V(\hat{b}_1) & \cdots & \text{cov}(\hat{b}_1, \hat{b}_{k-1}) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(\hat{b}_{k-1}, \hat{b}_0) & \text{cov}(\hat{b}_{k-1}, \hat{b}_1) & \cdots & V(\hat{b}_{k-1}) \end{pmatrix} \quad (13)$$

And notice that, for a non-random matrix A: $E(AZ) = A\,E(Z)$ but $V(AZ) = A\,V(Z)\,A'$

Usual assumptions of the model

These are the same as for the simple model, but generalized to any number of explanatory variables:

1. $E(u|X) = 0$
2. X and u are uncorrelated
3. $\text{rank}(X) = k$ with $k < N$
4. $E(uu') = \sigma^2 I_N$
5. When $N \to \infty$, $\lim \frac{X'X}{N} = V_X$ where V_X is a finite non-singular matrix

(2): consider for now that X is deterministic; we'll consider the more general case later. (3): the rank of a matrix is the maximum number of linearly independent columns: no perfect multicollinearity. (4): homoskedasticity and no autocorrelation. (5): a non-singular matrix has an inverse; the variance of the X's is always non-zero.

Estimation process (1)

- OLS minimizes the sum of squared residuals, each one called $\hat{u}_i$
- Calling $\hat{u}$ the vector of residuals, this sum can be written as: $S = \sum \hat{u}_i^2 = \hat{u}'\hat{u}$
- The derivative of S with respect to the vector of parameters $\hat{b}$ should be zero, for this vector of parameters to give an optimum for S
- Plus, the second derivative should be positive for this optimum to be a minimum
- We just need to compute the derivatives of S with respect to every single $\hat{b}_j$, and set them to 0

The gradient

In fact, we take the derivative of a number with respect to a vector, and get the vector G of derivatives, called the *gradient*:

$$G = \partial S / \partial \hat{b} = \begin{pmatrix} \partial S / \partial \hat{b}_0 \\ \partial S / \partial \hat{b}_1 \\ \vdots \\ \partial S / \partial \hat{b}_{k-1} \end{pmatrix} \quad (14)$$

After some calculus, we get (proof below):

$$\hat{b} = (X'X)^{-1}X'y \quad (15)$$

Optimization process (1)

We know that since $\hat{u} = y - \hat{y} = y - X\hat{b}$, then:

$$\begin{aligned} S &= \hat{u}'\hat{u} \\ &= (y - X\hat{b})'(y - X\hat{b}) \\ &= (y' - \hat{b}'X')(y - X\hat{b}) \\ &= y'y - y'X\hat{b} - \hat{b}'X'y + \hat{b}'X'X\hat{b} \\ &= y'y - 2y'X\hat{b} + \hat{b}'X'X\hat{b} \end{aligned}$$

Indeed, $y'X\hat{b}$ and $\hat{b}'X'y$ are each other's transpose, but they both are only scalars (matrix of size one by one), and a scalar and its transpose are the same thing.

Optimization process (2)

We can now compute the gradient G:

$$\begin{aligned} G &= \frac{\partial S}{\partial \hat{b}} \\ &= \frac{\partial (y'y)}{\partial \hat{b}} - \frac{\partial (2y'X\hat{b})}{\partial \hat{b}} + \frac{\partial (\hat{b}'X'X\hat{b})}{\partial \hat{b}} \\ &= 0 - 2X'y + 2X'X\hat{b} \end{aligned}$$

Since y is not a function of $\hat{b}$. We look for an optimum, so we want $G = 0$, which implies that $\hat{b} = (X'X)^{-1}X'y$ (given that $(X'X)^{-1}$ exists, which follows from the fact that $\text{rank}(X) = k$).

Optimization process (3)

- We also need the second derivative of S to be positive, which in the matrix framework amounts to having the matrix of second derivatives *positive definite*
- This matrix of second derivatives is equal to $\frac{\partial(-2X'y+2X'X\hat{b})}{\partial\hat{b}'} = 2X'X$, which is indeed positive definite
- Reminder: a matrix A is positive definite if for any non-zero vector z of convenient size, $z'Az > 0$. So let's check this with our matrix $2X'X$
- We only need to check that $X'X$ is positive definite. For any non-zero vector z, $z'(X'X)z = (Xz)'(Xz)$: a sum of squares, equal to the squared norm (=length) of vector Xz, which is > 0 since $Xz \neq 0$ (because $\text{rank}(X) = k$)

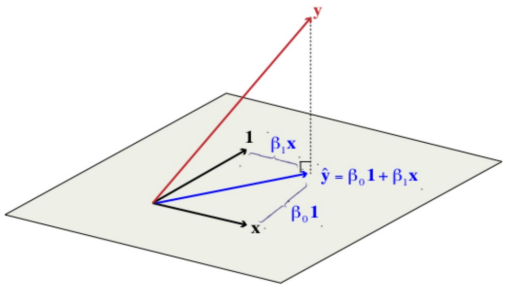
Geometrical interpretation of OLS (1)

See Wooldridge's *Introductory Econometrics*, Appendix E

- In the minimization process, we got to a point where:
  $G = -2X'y + 2X'X\hat{b} = 0$
- Rewritten:
  $X'y - X'X\hat{b} = X'(y - X\hat{b}) = X'(y - \hat{y}) = X'\hat{u} = 0$
- Which means that $\hat{u}$ should be perpendicular to every column vector of matrix X, i.e. perpendicular to the vector space spanned by the column vectors of X
- Condition $X'\hat{u} = 0$ is called the *system of normal equations* (*normal* also means perpendicular)

Illustration

Assume variable y is explained by 2 variables, a constant and x:
$y = \beta_0 + \beta_1 x + u$
OLS will find parameters $\hat{\beta}_0$ and $\hat{\beta}_1$ such that the model prediction $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ is as close as possible to the true y.



The vertical dashed line is vector $\hat{u}$ such that $y = \hat{y} + \hat{u}$. So we may find $\hat{\beta}$ starting directly from $X'\hat{u} = 0$ (normal equations)
Source: `https://sakai.unc.edu/`

Geometrical interpretation of OLS (2)

- Notice that $\hat{y} = X\hat{b} = X(X'X)^{-1}X'y = P_X y$
- Where: $P_X = X(X'X)^{-1}X'$
- Matrix P_X is in fact the general form of a projection matrix, which projects orthogonally on $L(X)$ (vector space spanned by the columns of X)
- Prediction $\hat{y}$ is the orthogonal projection of y on $L(X)$
- Let's call $M_X = I - P_X$, with I the identity matrix of convenient size
- M_X is a sort of complement to P_X
- M_X projects orthogonally on $L^{\perp}(X)$, the vector space orthogonal to $L(X)$
- We can see that the residual is $\hat{u} = M_X y$
- Notice too that $y = \hat{y} + \hat{u} = P_X y + M_X y$: y can be split into 2 orthogonal parts

Projectors

For any orthogonal projection matrix P_X, we have:

- $P_X P_X = P_X$: P_X is idempotent
- $P'_X = P_X$: P_X is symmetric

And some additional properties: $P_X + M_X = I$; $P_X M_X = 0$; $P_X X = X$; $M_X X = 0$. The eigenvalues of a projection matrix are equal either to 1 (corresponding to the vector space it projects onto) or 0 (corresponding to its orthogonal complement). For instance, P_X has k eigenvalues equal to 1 and $N - k$ eigenvalues equal to 0. (*From a geometrical point of view, OLS estimation is simply a projection.*)

Consequence: analysis of variance and R^2

- Vector y can be split in two orthogonal parts: $y = \hat{y} + \hat{u}$
- Thanks to the Pythagorean theorem (proof below), if the model has a constant:
- $\hat{V}(y) = \hat{V}(\hat{y}) + \hat{V}(\hat{u})$
- The coefficient of determination is: $R^2 = \hat{V}(\hat{y})/\hat{V}(y)$
- R^2 represents the percentage of variance explained by the model
- But adding a variable, even if it is irrelevant, will artificially increase the R^2
- For models with a different number of explanatory variables, only the $\bar{R}^2$'s can be compared
- Adjusted R^2: $\bar{R}^2 = 1 - \frac{N-1}{N-k}(1 - R^2)$ (warning: can be negative ...)

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Proof

- Call e_N the size N vector of 1's (the constant in the model), and subtract $\bar{y}e_N$ on both sides of $y = \hat{y} + \hat{u}$: $\quad y - \bar{y}e_N = (\hat{y} - \bar{y}e_N) + \hat{u}$
- $\hat{y} - \bar{y}e_N$ and $\hat{u}$ are orthogonal because $\hat{y} \perp \hat{u}$ and $e_N \perp \hat{u}$ (since e_N is a column of X and $X'\hat{u} = 0$)
- So we can apply the Pythagorean theorem:

$$\|y - \bar{y}e_N\|^2 = \|\hat{y} - \bar{y}e_N\|^2 + \|\hat{u}\|^2$$

- Since $\hat{u}$ averages to 0, the mean of $\hat{y}$ is $\bar{y}$, so this is equivalent to:

$$\sum(y_i - \bar{y})^2 = \sum(\hat{y}_i - \bar{\hat{y}})^2 + \sum(\hat{u}_i - \bar{\hat{u}})^2$$

- Dividing by N: $\hat{V}(y) = \hat{V}(\hat{y}) + \hat{V}(\hat{u})$

The R^2

$$R^2 = \frac{\sum(\hat{y}_i - \bar{y})^2}{\sum(y_i - \bar{y})^2} = \frac{\text{Explained sum of squares}}{\text{Total sum of squares}} = \frac{\hat{V}(\hat{y})}{\hat{V}(y)}$$
$$= 1 - \frac{\sum \hat{u}_i^2}{\sum(y_i - \bar{y})^2} = 1 - \frac{\text{Sum of squared residuals}}{\text{Total sum of squares}} = 1 - \frac{\hat{V}(\hat{u})}{\hat{V}(y)}$$

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Interpretations of the R^2

- R^2 represents the percentage of variance explained by the model
- It thus represents how well the model fits the data
- If there is a true relationship between the variables *but* this relationship is in fact non-linear, the R^2 will be rather low
- The R^2 is the square of the *multiple correlation coefficient* between y and the X's
- The R^2 is equal to the square of the linear correlation coefficient between y and its prediction $\hat{y}$
- If there is no constant in the model, the usual R^2 is meaningless (it can even be negative): statsmodels then reports an "uncentered" R^2
- The R^2 is not enough to assess the relevance of a regression: we'll need statistical tests (see later)

Remarks (1)

- R^2 and adjusted R^2 are valid only when comparing models that have the same dependent variable
- So they are inappropriate to compare 2 models with y and $\log(y)$ as the dependent variable
- Why: R^2 is a percentage of the variance of the outcome, so we can't compare percentages of different things
- See the PE test (MacKinnon, White and Davidson, 1983) to compare models in logs vs. levels (see Verbeek)

Remarks (2)

What happens to the R^2 if we add an irrelevant explanatory variable to the model?

- $\hat{y}$ is the orthogonal projection of y on $L(X)$
- If X is "larger" because we have one more explanatory variable, then $\hat{y}$ cannot be worse than before
- Either there is no improvement to the projection, or there is a small improvement due to randomness
- So the R^2 never decreases: it can only increase
- So don't rely on the R^2 only to judge a model